Exercise: Under "Case 3" it is easier to show that s^2 is an unbiased estimator of σ^2, and its also easy to derive the variance of this estimator. Do these now.

$$E(s^2) = E\left(\left(\frac{\sigma^2}{n-1}\right)\left(\frac{n-1}{\sigma^2}\right)s^2\right) = \frac{\sigma^2}{n-1} E\left(\frac{n-1}{\sigma^2} s^2\right) = \left(\frac{\sigma^2}{n-1}\right)(n-1) = \sigma^2$$

$$V(s^2) = V\left(\left(\frac{\sigma^2}{n-1}\right)\left(\frac{n-1}{\sigma^2}\right)s^2\right) = \frac{\sigma^4}{(n-1)^2} V\left(\frac{n-1}{\sigma^2} s^2\right) = 2\sigma^4/(n-1)$$

So, SE of s^2 is $\sqrt{\frac{2\sigma^4}{(n-1)}}$

Confidence Intervals

The already-mentioned standard error (SE) is a useful tool for quantifying error in an estimate, but a common alternative is the use of confidence intervals for unknown parameters.

Formally, we call (L, U) a $100(1-\alpha)\%$ confidence interval for θ if

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Hence, if want a "95% confidence interval," set $\alpha = 0.05$. Other common choices are 90% and 99% confidence intervals ($\alpha = 0.10$ and $\alpha = 0.01$, respectively).

An appropriate statement regarding a confidence interval is, for example, "I am 95% confident that the true value of θ is between 10.6 and 11.7."

It is **not** appropriate to say that "The probability that θ is between 10.6 and 11.7 is 0.95." This is incorrect because there is nothing random in that statement: θ, 10.6 and 11.7 are all constants.

The randomness is in the procedure itself. If one constructs 95% confidence intervals in a wide range of situations, then 95% of those intervals will cover the true value of the parameter.

There is an alternative approach to statistical inference, Bayesian inference, in which parameters are treated as random variables. More on that later.

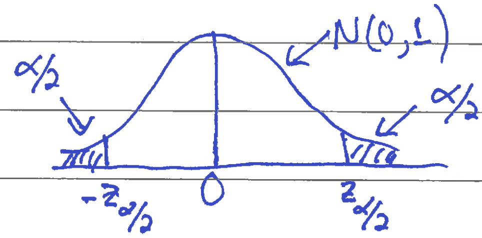
Exercise: Use our "classic results" above for "Case 3" to derive confidence intervals for μ and σ^2 in that situation. For estimating μ, onsider both the instances where σ^2 is known and unknown.

Recall: Case 3 implies $X_1, X_2, \ldots, X_n$ iid $N(\mu, \sigma^2)$

1 CI for μ, σ^2 is known

know that $\dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

So, $P\left(-z_{\alpha/2} \le \dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right) = 1-\alpha$



$z_{\alpha/2} = \text{qnorm}(1-\alpha/2)$

Rearrange so that μ is in the center alone:

$$P\Big(\underbrace{\bar{X} - z_{\alpha/2}\,\sigma/\sqrt{n}}_{L} \le \mu \le \underbrace{\bar{X} + z_{\alpha/2}\,\sigma/\sqrt{n}}_{U}\Big) = 1-\alpha$$

So, $\left(\bar{X} - z_{\alpha/2}\,\sigma/\sqrt{n},\ \bar{X} + z_{\alpha/2}\,\sigma/\sqrt{n}\right)$

is a $100(1-\alpha)\%$ CI for μ

Or: $\bar{X} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$

(2) CI for μ, σ unknown

know $\dfrac{\bar{X}-\mu}{s/\sqrt{n}} \sim t_{n-1}$

So, $P\left(-t_{\alpha/2,n-1} \le \dfrac{\bar{X}-\mu}{s/\sqrt{n}} \le t_{\alpha/2,n-1}\right) = 1-\alpha$

where $t_{\alpha/2,n-1} = qt(1-\alpha/2, n-1)$

Find

$$P\left(\bar{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \bar{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}\right) = 1-\alpha$$

So, $\bar{X} \pm t_{\alpha/2,n-1}\dfrac{s}{\sqrt{n}}$ is a $100(1-\alpha)\%$ CI for μ

Note that $t_{\alpha/2,n-1} > z_{\alpha/2}$

③ CI for σ^2

know $(n-1)s^2/\sigma^2 \sim$ chi-squared dist. with $n-1$ d.f.

Could find a, b such that

$$P\left(a \le \frac{(n-1)s^2}{\sigma^2} \le b\right) = 1-\alpha$$

i.e. $a = \text{qchisq}(\alpha/2, n-1)$ $\quad b = \text{qchisq}(1-\alpha/2, n-1)$

$$P\left(\frac{(n-1)s^2}{b} \le \sigma^2 \le \frac{(n-1)s^2}{a}\right) = 1-\alpha$$

So, $\left(\frac{(n-1)s^2}{b}, \frac{(n-1)s^2}{a}\right)$ is a $100(1-\alpha)\%$ CI for σ^2

The next slides present four important confidence intervals that are commonly used.

Confidence Interval for Population Mean, μ, when n is large ($n \geq 30$)

Assume that the sample is i.i.d. from a distribution with mean μ and variance σ^2, with both μ and σ^2 unknown. Then, a $100(1-\alpha)\%$ confidence interval for μ is

$$\overline{X} \pm z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

where z_a is such that $P(Z > z_a) = a$ when Z has the standard normal distribution. In R notation, z_a equals `qnorm(1-a)`.

Confidence Interval for Population Mean, μ, when n is small $(n < 30)$

Assume that the sample is i.i.d. from the normal distribution with mean μ and variance σ^2, with both μ and σ^2 unknown. Then, a $100(1-\alpha)\%$ confidence interval for μ is

$$\overline{X} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$$

where $t_{a,v}$ is such that $P(T > t_{a,v}) = a$ when T has the t-distribution with v degrees of freedom. In R notation, $t_{a,v}$ equals `qt(1-a,v)`.

Confidence Interval for Population Variance, σ^2

Assume that the sample is i.i.d. from the normal distribution with mean μ and variance σ^2, with both μ and σ^2 unknown. Then, a $100(1-\alpha)\%$ confidence interval for σ^2 is

$$\left(\frac{(n-1)\,s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)\,s^2}{\chi^2_{(1-\alpha/2),n-1}}\right)$$

where $\chi^2_{a,v}$ is such that $P(U > \chi^2_{a,v}) = a$ when U has the chi-squared distribution with v degrees of freedom. In R notation, $\chi^2_{a,v}$ equals `qchisq(1-a,v)`.

Confidence Interval for Population Proportion, p

Assume that n is large enough that you are confident that $np \geq 5$ and $n(1-p) \geq 5$. Let $\hat{p} = X/n$ where X is binomial(n, p), and p is unknown. Then, a $100(1-\alpha)\%$ confidence interval for p is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where z_a is such that $P(Z > z_a) = a$ when Z has the standard normal distribution. In R notation, z_a equals `qnorm(1-a)`.

Exercise: A sample of size 50 is drawn from a population with unknown mean and variance. It holds that $\bar{x} = 14.6$ and $s = 1.3$. Estimate the population mean. State both the standard error for the estimator and a 90% confidence interval for μ.

Best estimate of μ is $\bar{X} = 14.6$

SE of $\bar{X}$ is $\sqrt{V(\bar{X})} = \sqrt{\sigma^2/n} \approx s/\sqrt{n} = 1.3/\sqrt{50}$

90% CI for μ is $\bar{X} \pm z_{\alpha/2}\left(s/\sqrt{n}\right)$

$14.6 \pm (z_{.05})\left(1.3/\sqrt{50}\right)$

$14.6 \pm (1.64)\left(1.3/\sqrt{50}\right) = (14.30, 14.90)$

"I am 90% confident that μ is between 14.3 and 14.9."

Note that when $\alpha = 0.05$, then $z_{\alpha/2} \approx t_{\alpha/2, n-1} \approx 2$

Part 11: Maximum Likelihood Estimation

Summary

Maximum likeihood estimation (MLE) is a powerful, general approach to constructing estimators.

We will begin the explanations with simple, one-dimensional examples, but the concepts extend to complex, high-dimensional models in a natural manner.

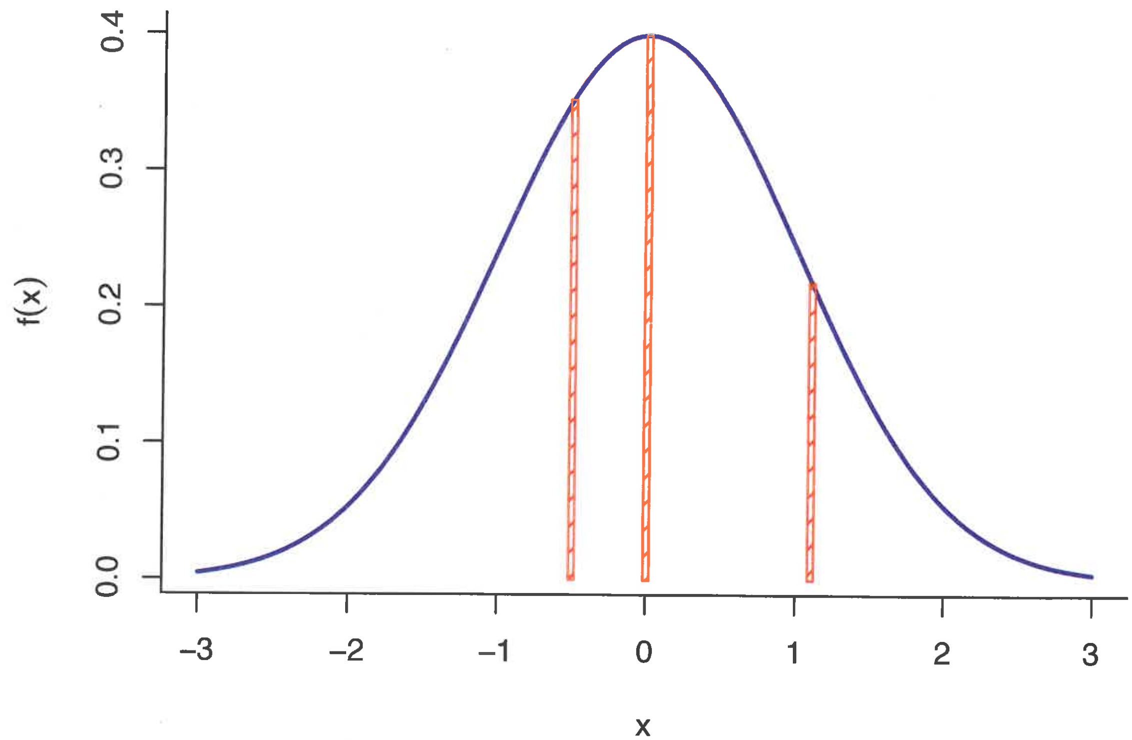
Importantly, there is also a general way of calculating standard errors of MLEs, and, under a wide range of conditions, the MLE is approximately normally distributed. This makes inference with MLEs straightforward.

Basic Setup

Recall that if X is a continuous random variable with density f_X, that does **not** mean that $P(X = a) = f_X(a)$, i.e., the density does **not** return probabilities. Indeed, $P(X = a) = 0$ for all a.

It is true that the density is higher in regions where there is the most probability: For fixed, small $\epsilon > 0$, if we want to find a to maximize $P(a - \epsilon \leq X \leq a + \epsilon)$, then we would look for the a that maximizes $f_X(a)$.

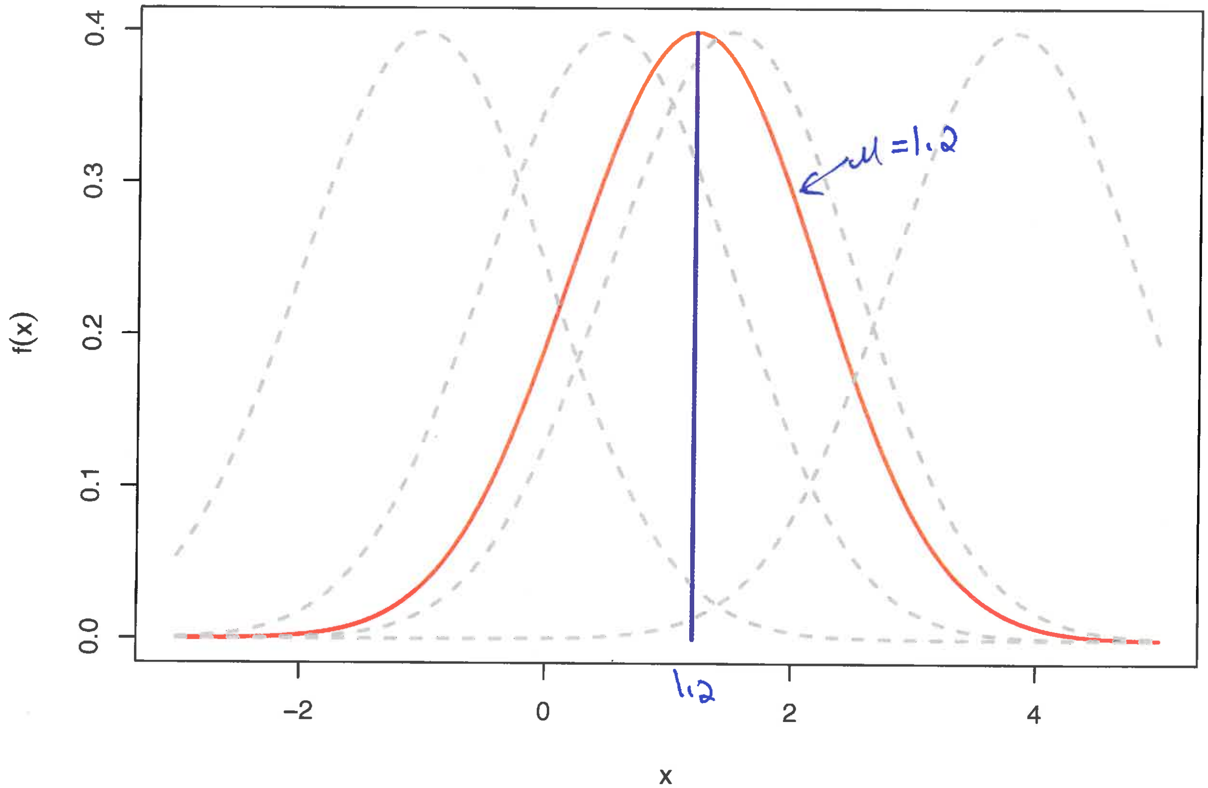
See the figure on the following slide. The standard normal density is depicted. The most likely values are close to zero, where the density is maximized.



As previously stated, we use probability distributions such as the normal to "model" the real processes that generate data. So, if we assume that our observable data is drawn from the standard normal distribution, the density above specifies where we are more or less likely to see our data value.

We can turn this idea around: Suppose that we observe data $x = 1.2$, and we are willing to assume that it came from a normal distribution with mean μ and variance one. What value of μ makes the observation $x = 1.2$ the most likely?

See the figure on the following slide. Different choices of μ are displayed. It's not difficult to conclude that when $\mu = 1.2$, the likelihood is maximized.



The likelihood function is constructed by calculating the distribution for the data, and then **holding the data constant** and allowing the parameter(s) to vary.

In the simple example above, we assume that we observe a single data point x that was generated from a normal distribution with mean μ and variance one. Hence, the distribution is

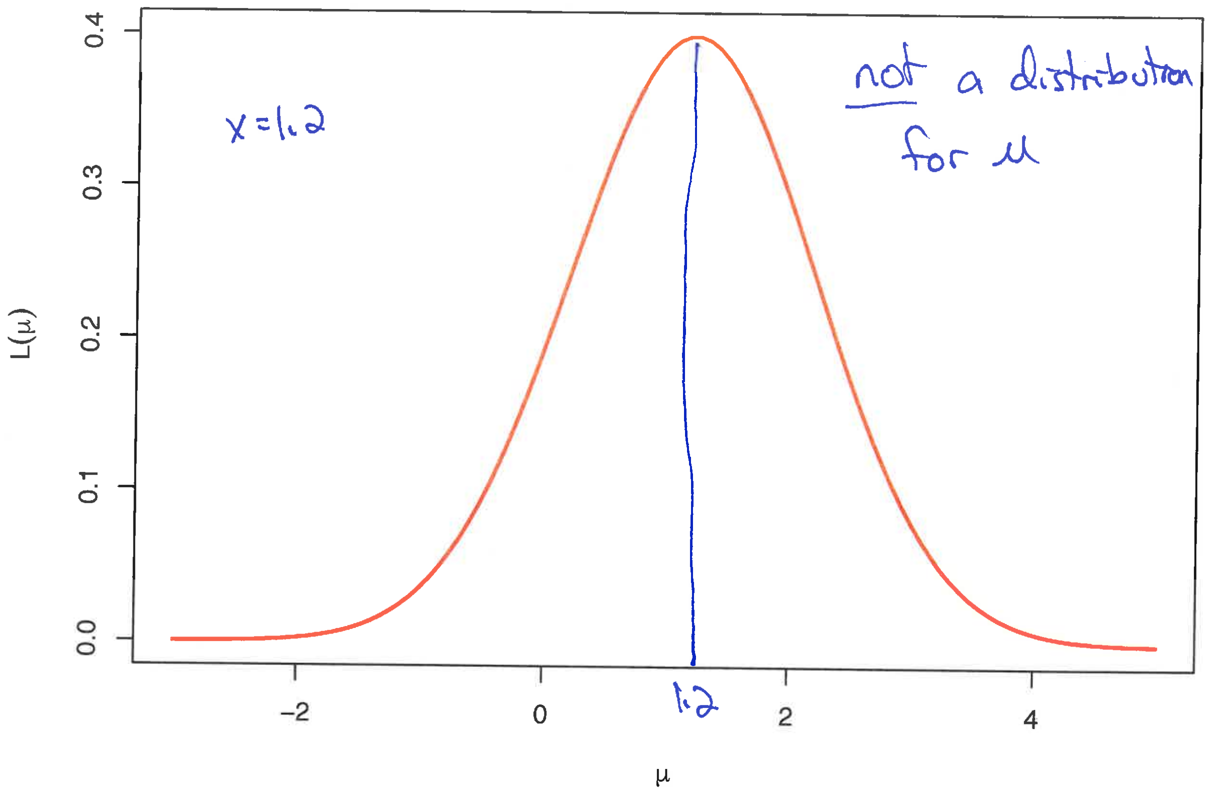
$$f(x;\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right)$$

If we fix the data $x = 1.2$, and think of this as a function of μ, the we arrive at our likelihood function:

$$L(\mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1.2-\mu)^2}{2}\right)$$

<u>not</u> a density

$\int L(\mu)$



The maximum likelihod estimator for the parameter is the choice of the parameter which maximizes the likelihood function.

Heuristically, we ask: Among all possible choices for the parameter, which one makes our observed data the most likely?

It is worth stressing that the likelihood function is **not** itself a probability distribution. It is **not** specifying the relative "probabilities" for different values of the parameter.

Remember that in the frequentist approach, the unknown parameter is a constant, and not random.

Exercise: An urn contains black and white balls. All that we know is that the balls have a color ratio of 1 to 3. (We do not know if there are more white or more black balls.) We draw three balls **with replacement** from the urn. Let X equal the number of black draws. Derive the likelihood function and the MLE as a function of X.

$\theta = P(\text{single draw is black})$, θ is either $1/4$ or $3/4$

$X \sim \text{binomial}(n=3, \theta)$

	$x=0$	$x=1$	$x=2$	$x=3$
$\theta = 1/4$	$27/64$	$27/64$	$9/64$	$1/64$
$\theta = 3/4$	$1/64$	$9/64$	$27/64$	$27/64$

$$\hat{\theta}_{MLE} = \begin{cases} 1/4, & x = 0,1 \\ 3/4, & x = 2,3 \end{cases}$$

Exercise: Now suppose that θ equals the proportion of balls in the urn which are black. We know that $0 < \theta < 1$. We draw ten times with replacement, and count seven black balls and three white balls. Derive the MLE for θ.

$X = \#$ of black balls. Observe $x = 7$

$$f_X(x;\theta) = \binom{10}{x} \theta^x (1-\theta)^{10-x}$$

The likelihood function is $L(\theta) = \binom{10}{7} \theta^7 (1-\theta)^3$

$$\frac{\partial L(\theta)}{\partial \theta} = \binom{10}{7} \left[7\theta^6 (1-\theta)^3 - 3\theta^7 (1-\theta)^2 \right]$$

$$\frac{\partial L(\theta)}{\partial \theta} = 0 \Rightarrow \hat{\theta}_{MLE} = 0.7$$

should check second deriv. to verify found max

The preceding examples involved a single, observable random variable. In most cases, we have a sequence of random variables in the model.

A common case is where $X_1, X_2, \ldots, X_n$ are i.i.d. with density $f_X(x;\theta)$, where θ is the parameter to be estimated. In this case, the joint distribution of the random variables is

$$\prod_{i=1}^{n} f_{X_i}(x_i;\theta)$$

and the likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f_{X_i}(x_i;\theta)$$

Finding the MLE Analytically

In some situations the MLE can be derived analytically, usually via calculus, but sometimes via direct inspection of the likelihood.

One "trick" that is often very helpful: The likelihood function is positive at its maximum. Hence, the likelihood is maximized at the same θ as does the logarithm of the likelihood.

Hence, we often work with the log likelihood instead of the likelihood. This is often denoted $\ell(\theta) = \log L(\theta)$.

In the case where $X_1, X_2, \ldots, X_n$ are i.i.d. $f_X(x;\theta)$, then

$$\ell(\theta) = \log \prod_{i=1}^{n} f_{X_i}(x_i;\theta) = \sum_{i=1}^{n} \log f_{X_i}(x_i;\theta)$$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. with the Exponential(λ) distribution. Derive the MLE for λ.

$$f_X(x_i;\lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

$$L(\lambda) = \prod_{i=1}^{n} f_{X_i}(x_i;\lambda)$$

$$= \prod_{i=1}^{n} \lambda e^{-\lambda x_i}, \quad x_i > 0$$

$$= \lambda^n e^{-\lambda \sum x_i}, \quad x_i > 0$$

$$\log L(\lambda) = n \log \lambda - \lambda \sum x_i$$

$$\frac{\partial \log L(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum x_i$$

Set equal to 0, $\hat{\lambda}_{MLE} = 1/\bar{x}$

$$\boxed{\begin{aligned} \log a^b &= b \log a \\ \log ab &= \log a + \log b \\ \log e^a &= a \end{aligned}}$$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. with the Beta$(\alpha, 1)$ distribution. Derive the MLE for α. Compare this with the MOM estimator.

$$f_X(x;\alpha) = \alpha x^{\alpha-1}, \quad 0 < x < 1$$

$$L(\alpha) = \prod f_X(x_i;\alpha) = \alpha^n \left(\prod x_i\right)^{\alpha-1}, \quad 0 < x_i < 1$$

$$\log L(\alpha) = n \log \alpha + (\alpha - 1) \log\left(\prod x_i\right)$$

$$\frac{\partial \log L(\alpha)}{\partial \alpha} = \frac{n}{\alpha} + \log\left(\prod x_i\right)$$

Set equal to 0, find $\hat{\alpha}_{MLE} = -\dfrac{n}{\log(\prod x_i)} = -\dfrac{n}{\sum \log x_i}$

$E(X) = \dfrac{\alpha}{\alpha+1}$, $m_1 = \bar{X}$ $\quad \hat{\alpha}_{mom} = \dfrac{\bar{X}}{1-\bar{X}}$